The Machine Learning Algorithm Map

Ten core algorithms sorted by the data they work on and by how they learn: with labels, without labels, or from the data itself.

WHERE THEY SIT

Artificial Intelligence
Machines doing tasks that seem intelligent

Machine Learning
Learning patterns from data (regression, trees, SVM, K-means, PCA)

Deep Learning
Many-layer neural networks (CNN, Transformer, diffusion)

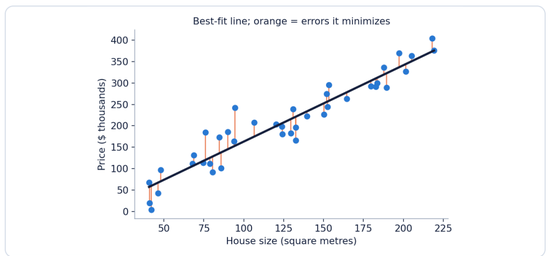
Generative AI
Creates new text and images (Transformer/GPT, diffusion)

How it learns: SUPERVISED uses labels · SELF-SUPERVISED makes its own labels · UNSUPERVISED finds patterns, no labels

Tags = also used in: TABULAR · VISION · LANGUAGE · GEN AI

Tabular Data

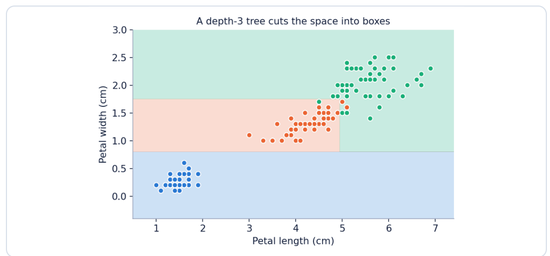
Rows and columns: sales, patients, customers, transactions



Linear Regression — SUPERVISED

How: fits the straight line that has the smallest total error, then uses it to predict a number.

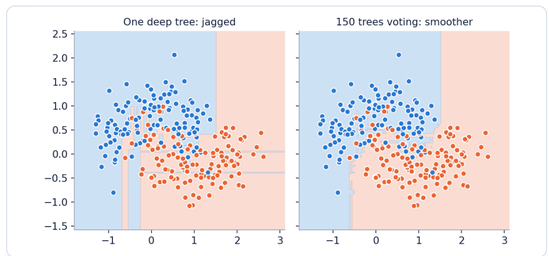
Use: predicting house prices.



Decision Tree — SUPERVISED

How: asks a chain of yes/no questions about the features until it reaches an answer.

Use: loan approval.

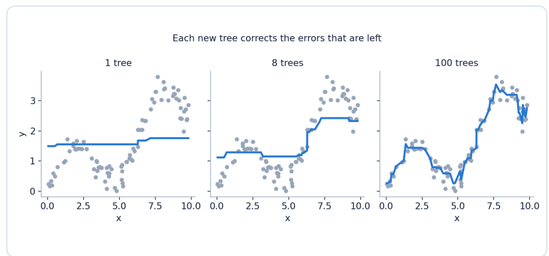


Random Forest — SUPERVISED

↳ many Decision Trees

How: grows many different trees on random samples and lets them vote.

Use: fraud detection.



Gradient Boosting — SUPERVISED

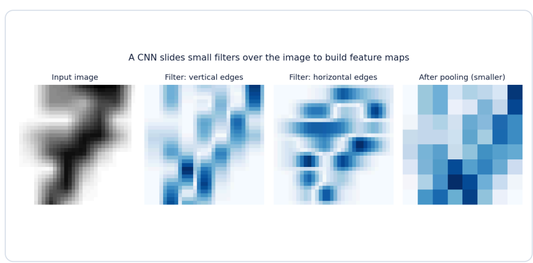
↳ builds on Decision Trees

How: adds small trees one at a time, each fixing the errors the last ones left.

Use: customer churn prediction.

Computer Vision

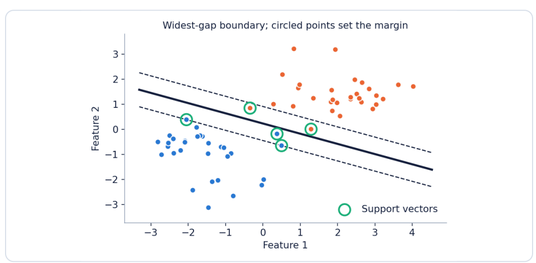
Photos, video and scans



CNN — SUPERVISED

How: slides small filters over an image to find edges, then shapes, then whole objects.

Use: image recognition.



SVM — SUPERVISED

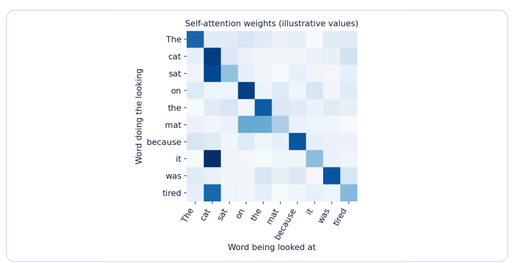
TABULAR · LANGUAGE

How: draws the boundary that leaves the widest gap between two classes.

Use: image classification.

Natural Language

Emails, reviews, documents, chat



Transformer — SELF-SUPERVISED

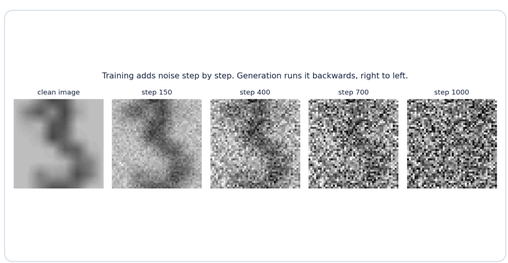
GEN AI

How: every word looks at every other word to decide what matters. GPT predicts the next word, BERT fills in hidden ones.

Use: text generation and classification.

Generative AI

Creates new images, text and designs



Diffusion Model — SELF-SUPERVISED

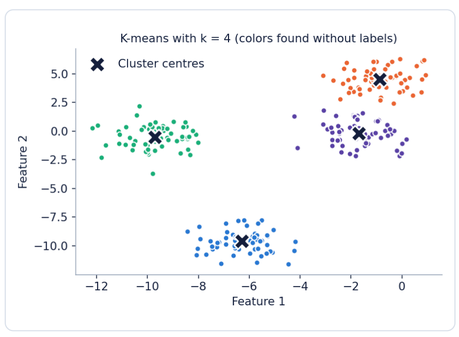
VISION

How: learns to remove noise, then starts from pure noise and cleans it up, guided by a prompt.

Use: image generation (Stable Diffusion).

Cross-domain tools that need no labels

These work on tables, images and text alike, so they sit under every column

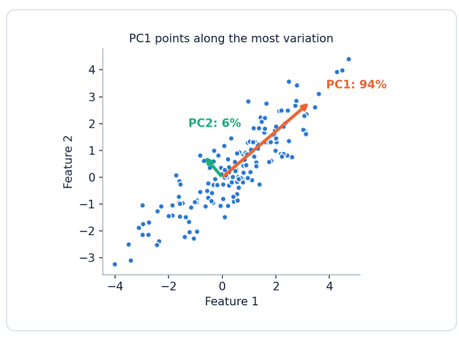


K-Means Clustering — UNSUPERVISED

TABULAR · VISION

How: places k centres, assigns each point to the nearest one, moves each centre to the middle of its group, and repeats until nothing changes.

Use: customer segmentation.



PCA — UNSUPERVISED

TABULAR · VISION

How: finds the few directions that hold most of the variation and squeezes many columns down to those.

Use: reducing data dimensions.

Which one should I reach for?

A table of numbers
Start with Random Forest or Gradient Boosting. Linear Regression for a simple, explainable number.

Images
CNN is the standard. SVM can work when you have little data and good hand-made features.

Text
A Transformer: BERT-style models to understand it, GPT-style models to write it.

Make something new
Diffusion for images, a GPT-style Transformer for text.

No labels yet
K-Means to find groups, PCA to simplify the data first.

Deepthi Pinninti | AIML Portfolio Artifact, Workshop 1.5

Diagrams use real example data, except the attention map, which is illustrative. CNN is usually supervised; Transformers are pre-trained self-supervised, then often fine-tuned with labels.